1. Two parallel conducting plates are a distance d apart and have a potential difference V between them. A uniform magnetic field B is also present in the region between the plates along a direction parallel to the plates. Particles of charge +q and mass m are released from rest in the crossed E and B fields from the surface of one plate. Find the trajectory of the particles.

Sol: First let us set up the coordinate system with its z-axis along the direction to $\vec{B}$ so that $\vec{B} = B\hat{k}$ and $\vec{E} = E\hat{\jmath} = \frac{V}{d}\hat{\jmath}$. $\vec{B}$ is out of this page as shown in fig 1.

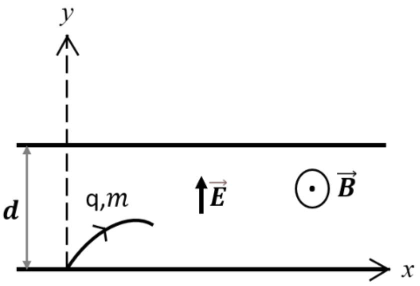

Fig. 1: Schematic of coordinate system indicating the direction of electric and magnetic field.

The electric force is always along the y- direction and the magnetic force being perpendicular to $B\hat{k}$ will always be in the $x-y$ plane which implies that the particle can never come out of the $x-y$ plane, and the most general expression for its velocity is

$\vec{v} = \dot{x}\hat{\imath} + \dot{y}\hat{\jmath} \quad [\, \dot{x} = \frac{dx}{dt} , \dot{y} = \frac{dy}{dt}]$

Now, Lorentz force, $\quad \vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

$$\Rightarrow \quad m(\ddot{x}\hat{\imath} + \ddot{y}\hat{\jmath}) = qE\hat{\jmath} + q(\dot{x}\hat{\imath} + \dot{y}\hat{\jmath}) \times \mathrm{B}\hat{k}$$

Let us define $\omega = \frac{qB}{m}$, cyclotron frequency

Then we have

$$\ddot{x} = \omega\dot{y}$$

$$\ddot{y} = -\omega\dot{x} + \frac{qE}{m}$$

Now $\quad \dddot{x} = \omega\ddot{y} = -\omega^2\dot{x} + \frac{\omega qE}{m}$

and $\quad \dddot{y} = -\omega\ddot{x} = -\omega^2\dot{y} \quad \Rightarrow \quad \frac{d^2\dot{y}}{dt^2} = -\omega^2\dot{y}$

which leads to the solution, $\quad \dot{y} = A\cos\omega t + C\sin\omega t$

Therefore, $\quad \boldsymbol{y = \frac{A}{\omega}\sin\omega t - \frac{C}{\omega}\cos\omega t + D} \quad$ where D is integration constant

Then $\quad \ddot{y} = -A\omega\sin\omega t + C\omega\cos\omega t$

At $\; t = 0, \; y = 0, \dot{y} = 0$ and $\ddot{y} = \frac{qE}{m}$

Using initial condition, we have $A = 0\, , C = \frac{qE}{\omega m}$ and $D = \frac{qE}{\omega^2 m} = \frac{E}{\omega B}$ using $\omega = \frac{qB}{m}$

Therefore, we have,

$$y = \frac{E}{\omega B}(1 - cos\omega t)$$

Now using the equation $\ddot{x} = \omega \dot{y}$

we have, $$\ddot{x} = \omega \frac{E}{\omega B}(\omega\, sin\omega t) = \frac{E}{B}\, \omega\, sin\omega t$$

Therefore, $\dot{x} = -\frac{E}{B}\, \omega\, \frac{cos\omega t}{\omega} + G = -\frac{E}{B} cos\omega t + G$ where G is integration constant

Finally, we get $\boldsymbol{x = -\frac{E}{B}\, \frac{sin\omega t}{\omega} + Gt + M}$

At $t = 0,\ x = 0,\ \dot{x} = 0$ and $\ddot{x} = 0$

So we have, $G = \frac{E}{B}$, $M = 0$

Thus, $\boldsymbol{x = -\frac{E}{B}\, \frac{sin\omega t}{\omega} + \frac{E}{B} t}$

At the end we get $$\boldsymbol{x = \frac{E}{\omega B}\,(\omega t - sin\omega t)}$$

$$\boldsymbol{y = \frac{E}{\omega B}\,(1 - cos\omega t)}$$

These equations are the parametric equation of a cycloid. So the trajectory of the particle is shown in fig 2 below.

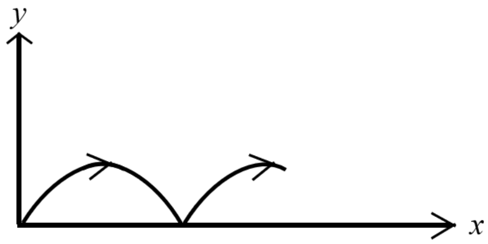

Fig. 2: Schematic of trajectory of the particle

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2. A loop carrying current is shaped like a regular plane polygon of 2n sides, the distance between parallel sides being 2a. Find the magnetic field at the centre of the loop.

Sol: Let us consider just one of the sides as shown in fig below. The side is subtending an angle $\frac{2\pi}{2n}$ at the centre, O .

From the figure, $S = a\, tan\,\theta \ \Rightarrow\ dS = asec^2\theta$

Also, $a = rcos\theta$

Now,

$$B = 2n\, \frac{\mu_0 i}{4\pi} \int \frac{\sin\alpha}{r^2}\, dS = \frac{n\mu_0 i}{2\pi a} \int_{-\frac{\pi}{2n}}^{\frac{\pi}{2n}} cos\theta\, d\theta = \frac{n\mu_0 i}{2\pi a}\, 2\sin\frac{\pi}{2n}$$

Therefore, $B = \frac{n\mu_0 i}{\pi a} \sin\frac{\pi}{2n}$

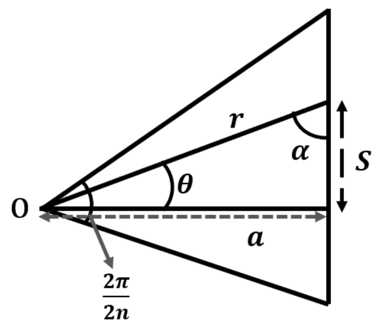
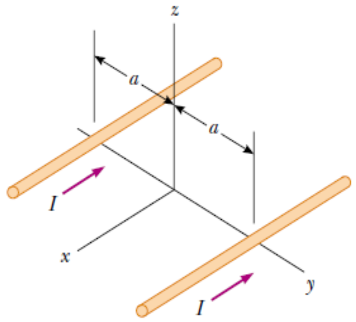


Fig.2 : Schematic of one of the sides of the polygon

3. In Figure below, both currents in the infinitely long wires are in the negative x direction. (a) Sketch the magnetic field pattern in the yz plane. (b) At what distance d along the z axis is the magnetic field a maximum?



Sol: (a)

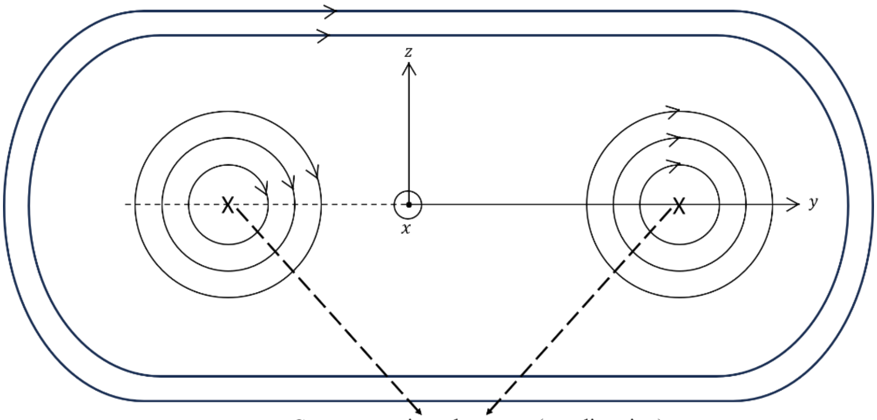


Fig. 3a

(b) At a point on the z-axis from each wire $B = \frac{\mu_0 I}{2\pi\sqrt{a^2+z^2}}$ and it is perpendicular to the line from to the wire as shown in fig 3b.

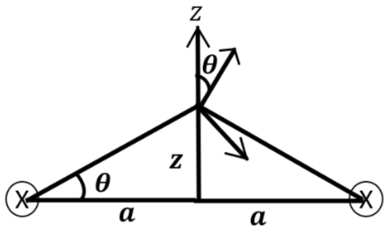


Fig. 3b

Here vertical components cancel while the horizontal components get added. Therefore, $B_y = 2\,\frac{\mu_0 I}{2\pi\sqrt{a^2+z^2}}\,sin\theta$

$$= \frac{\mu_0 I}{\pi\sqrt{a^2+z^2}}\,\frac{z}{\sqrt{a^2+z^2}}$$

$$= \frac{\mu_0 I z}{\pi\,(a^2+z^2)}$$

The condition for maximum is $\frac{dB_y}{dz} = 0$

$$\Rightarrow \frac{\mu_0 I z}{\pi\,(a^2+z^2)} - \frac{\mu_0 I z}{\pi\,(a^2+z^2)^2} \times 2z = 0$$

$$\Rightarrow z = a$$

Therefore, the field is maximum at $z = a$

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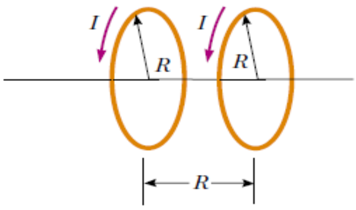
4. Two circular coils of radius R, each with N turns, are perpendicular to a common axis. The coil centers are a distance R apart. Each coil carries a steady current I in the same direction, as shown in Figure below.



(a) Show that the magnetic field on the axis at a distance x from the center of one coil is

$$B = \frac{N\mu_0 I R^2}{2}\left[\frac{1}{\left(R^2+x^2\right)^{3/2}} + \frac{1}{\left(2R^2+x^2-2Rx\right)^{3/2}}\right]$$

(b) Show that dB/dx and d^2B/dx^2 both are zero at the point midway between the coils. This means the magnetic field in the region midway between the coils is uniform.

Sol:

(a) Magnetic field at a distance x on the axis at a circular loop of radius R is given by (worked out in the lecture)

$$B_x = \frac{\mu_0 I R^2}{2\,(x^2+R^2)^{3/2}}$$

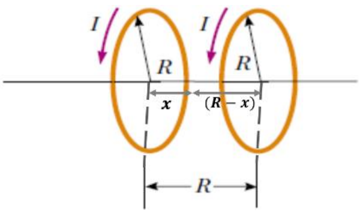


Fig. 4

Now each coil has N turns. So,

$$B = B_{1x} + B_{2x}$$

$$= \frac{N\mu_0 I R^2}{2}\left[\frac{1}{(x^2+R^2)^{\frac{3}{2}}} + \frac{1}{((R-x)^2+R^2)^{\frac{3}{2}}}\right]$$

$$= \frac{N\mu_0 I R^2}{2}\left[\frac{1}{(x^2+R^2)^{\frac{3}{2}}} + \frac{1}{(x^2+2R^2-2Rx)^{\frac{3}{2}}}\right]$$

(b). $$\frac{dB}{dx} = \frac{N\mu_0 I R^2}{2}\left[-\frac{3}{2}\frac{2x}{(x^2+R^2)^{\frac{5}{2}}} - \frac{3}{2}\frac{(2x-2R)}{(x^2+2R^2-2Rx)^{\frac{5}{2}}}\right]$$

$$= \frac{N\mu_0 I R^2}{2}\left[-\frac{3x}{(x^2+R^2)^{\frac{5}{2}}} - \frac{3(x-R)}{(x^2+2R^2-2Rx)^{\frac{5}{2}}}\right]$$

Now, $\frac{dB}{dx} = 0 \Rightarrow \frac{x}{(x^2+R^2)^{\frac{5}{2}}} - \frac{(R-x)}{(x^2+2R^2-2Rx)^{\frac{5}{2}}}$

A little inspection easily leads to $\boldsymbol{x} = \frac{\boldsymbol{R}}{\mathbf{2}}$

Again, $\frac{d^2B}{dx^2} = \frac{N\mu_0 I R^2}{2}\left[-\frac{3x}{(x^2+R^2)^{\frac{5}{2}}} + \frac{5}{2}\frac{3x\,(2x)}{(x^2+2R^2-2Rx)^{\frac{7}{2}}} - \frac{3}{(x^2+2R^2-2Rx)^{\frac{5}{2}}} + \frac{5}{2}\frac{3(x-R)}{(x^2+2R^2-2Rx)^{\frac{7}{2}}}\right]$

Putting $x = \frac{R}{2}$ we get $\frac{d^2B}{dx^2} = 0$

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5. The surface current density in the xoy plane is given to be

$$\vec{K}(x,y) = K_0\left(\frac{x^2}{a^2}\hat{\imath} + \frac{y^2}{b^2}\hat{\jmath}\right)$$

Evaluate $\vec{\nabla}.\vec{K}$ and hence calculate the time rate of change of charge contained in the first quadrant of a circle of unit radius centred on the origin. Show explicitly that for this quadrant, the continuity equation is satisfied in its integral form.

Sol: $\vec{K} = K_0\left(\frac{x^2}{a^2}\,\hat{\imath} + \frac{y^2}{b^2}\,\hat{\jmath}\right)$

Then, $\vec{\nabla}.\vec{K} = 2K_0\left(\frac{x}{a^2} + \frac{y}{b^2}\right)$

Continuity equation in differential form, for surface current $\vec{K}$ is

$$\vec{\nabla}.\vec{K} = -\frac{\partial\sigma}{\partial t}$$

$\vec{K}$ is along $\hat{\imath}$ for all points along OP while it is along $\hat{\jmath}$ along QO as shown in fig 5. There is no charge flowing across OP & QO. That will leave the part PQ of the circle. As we have $d\vec{l} = \hat{r}dr + \hat{\theta}\,rd\theta$ $= \hat{\theta}\,d\theta$, since $r = 1$ for unit circle.

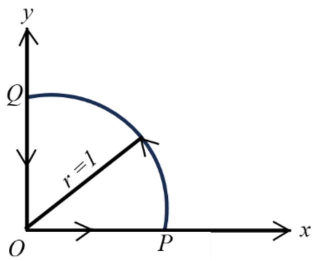

Fig. 5

Again, $\vec{K} = K_0\left(\frac{x^2}{a^2}\,\hat{\imath} + \frac{y^2}{b^2}\,\hat{\jmath}\right)$

$$= K_0\left[\frac{r^2cos^2\theta}{a^2}\left(\hat{r}\,cos\theta - \hat{\theta}\,sin\theta\right) + \frac{r^2sin^2\theta}{b^2}\left(\hat{r}\,sin\theta + \hat{\theta}\,cos\theta\right)\right]$$

This leads to $|\vec{K}\times d\vec{l}| = K_0\left[\frac{cos^3\theta}{a^2} + \frac{sin^3\theta}{b^2}\right]d\theta$

$$\Rightarrow \frac{\partial \sigma}{\partial t} = -2K_0 \left[\frac{x}{a^2} + \frac{y}{b^2}\right]$$

Changing to (r, θ) coordinate, $\frac{\partial \sigma}{\partial t} = -2K_0 \left[\frac{rcos\theta}{a^2} + \frac{rsin\theta}{b^2}\right]$

If q is the charge in the first quadrant of a circle of unit radius, then

$$\frac{\partial q}{\partial t} = \int_0^1 \int_0^{\frac{\pi}{2}} \frac{\partial \sigma}{\partial t}\, r\, dr\, d\theta$$

$$= -2K_0 \int_0^1 \int_0^{\frac{\pi}{2}} \left(\frac{rcos\theta}{a^2} + \frac{rsin\theta}{b^2}\right) r\, dr\, d\theta$$

$$= -2K_0 \left[\left(\frac{1}{a^2}\frac{r^3}{3}\right)\Big|_0^1 (sin\theta)\Big|_0^{\frac{\pi}{2}} - \left(\frac{1}{b^2}\frac{r^3}{3}\right)\Big|_0^1 (cos\theta)\Big|_0^{\frac{\pi}{2}}\right]$$

$$\Rightarrow \frac{\partial q}{\partial t} = -\frac{2K_0}{3}\left[\frac{1}{a^2} + \frac{1}{b^2}\right] \qquad (i)$$

Let us now evaluate the total charge q flowing at of the boundaries of the first quadrant of the unit circle. As we are interested in

$$\int_P^Q \left|\vec{K} \times d\vec{l}\right| = K_0 \int_0^{\frac{\pi}{2}} \left[\frac{cos^3\theta}{a^2} + \frac{sin^3\theta}{b^2}\right] d\theta = \frac{2K_0}{3}\left[\frac{1}{a^2} + \frac{1}{b^2}\right] \qquad (ii)$$

Equations (i) and (ii) explicitly show that the rate of loss of charge from boundaries of the quadrant is the same as the rate of depletion of charge within quadrant. The continuity equation is thus satisfied in its integral form.

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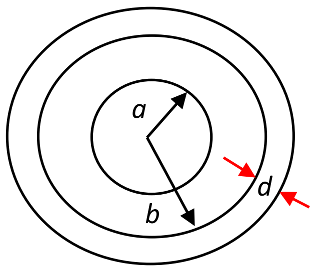
6. Consider an infinitely long transmission line consisting of two concentric cylinders having their axes along the z-axis. The cross-section of the transmission line is shown in figure below where z-axis is out of the page. The inner conductor has radius a and carries current I while the outer conductor has inner radius b and thickness d and carries return current –I. Determine everywhere, assuming that current is uniformly distributed in both conductors.



Sol: Let us apply Ampere's law due to symmetrical current distribution, along the Amperian path for each of the four possible regions:

$(i)\ 0 \le \rho \le a$ $\quad (ii)\ a \le \rho \le b$ $\quad (iii)\ b \le \rho \le b + d$ $\quad (iv)\ \rho \ge b + d$

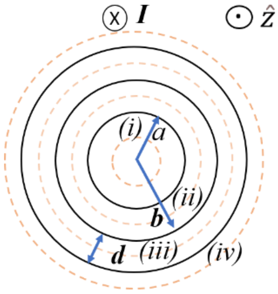


Fig. 6a

$(\boldsymbol{i})\ \mathbf{0} \le \boldsymbol{\rho} \le \boldsymbol{a}$

$\oint \vec{B}.d\hat{l} = \mu_0 I_{enc} = \mu_0 \oint \vec{j}.d\hat{s}$

$\vec{j} = \frac{I}{\pi a^2}\hat{z}$ and $d\hat{s} = \rho\, d\rho\, d\varphi\, \hat{z}$

$$I_{enc} = \oint \vec{j}.d\hat{s} = \frac{I}{\pi a^2} = \int_0^{\rho}\int_0^{2\pi} d\varphi \;\; \rho\, d\rho = \frac{I\rho^2}{a^2}$$

Thus, $B\, 2\pi\rho = \frac{\mu_0 I \rho^2}{a^2}$ which gives $\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0 I \rho}}{\boldsymbol{2\pi a^2}}\,\hat{\boldsymbol{\varphi}}$ for $0 \le \rho \le a$

$(\boldsymbol{ii})\ \boldsymbol{a} \le \boldsymbol{\rho} \le \boldsymbol{b}$

$\oint \vec{B}.d\hat{l} = \mu_0 I_{enc} = \mu_0 \mathrm{I}$

Thus, $B\, 2\pi\rho = \mu_0 \mathrm{I}$ which gives $\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0 I}}{2\pi\rho}\,\hat{\boldsymbol{\varphi}}$ for $a \le \rho \le b$

$(\boldsymbol{iii})\ \boldsymbol{b} \le \boldsymbol{\rho} \le \boldsymbol{b} + \boldsymbol{d}$

$\oint \vec{B}.d\hat{l} = \mu_0 I_{enc} \Rightarrow B\, 2\pi\rho = \mu_0 I_{enc}$

Here $I_{enc} = I + \int \vec{j}.d\hat{s}$

$\vec{j}$ in this case is the current density of the outer conductor and it is along $-\hat{z}$ direction. So,

$$\vec{j} = -\frac{I}{\pi[(b+d)^2 - b^2]}\hat{z}$$

Thus, $I_{enc} = I - \frac{I}{\pi[(b+d)^2 - b^2]}\int_0^{2\pi}\int_0^{\rho} \rho\, d\rho\, d\varphi = I\left[1 - \frac{\rho^2 - b^2}{d^2 + 2bd}\right]$

Hence, $\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0 I}}{2\pi\rho}\left[1 - \frac{\rho^2 - b^2}{d^2 + 2bd}\right]\hat{\boldsymbol{\varphi}}$ for $b \le \rho \le b + d$

$(\boldsymbol{iv})\ \boldsymbol{\rho} \ge \boldsymbol{b} + \boldsymbol{d}$

$$I_{enc} = I - I = 0$$

Hence, $\vec{\boldsymbol{B}} = \mathbf{0}$ for $\rho \ge b + d$

Finally, $\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0 I \rho}}{\boldsymbol{2\pi a^2}}\ \hat{\boldsymbol{\varphi}}$ for $0 \leq \rho \leq a$

$\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0 I}}{2\pi\rho}\ \hat{\boldsymbol{\varphi}}$ for $a \leq \rho \leq b$

$\vec{\boldsymbol{B}} = \frac{\boldsymbol{\mu_0 I}}{2\pi\rho}\left[1 - \frac{\rho^2 - b^2}{d^2 + 2bd}\right]\hat{\boldsymbol{\varphi}}$ for $b \leq \rho \leq b + d$

$\vec{\boldsymbol{B}} = \boldsymbol{0}$ for $\rho \geq b + d$

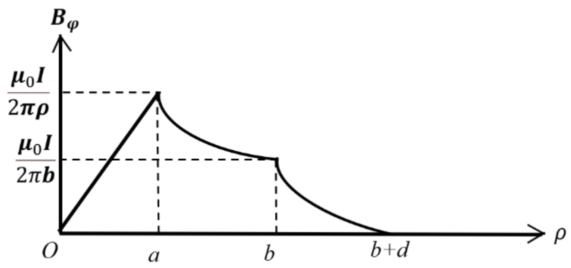


Fig. 6b